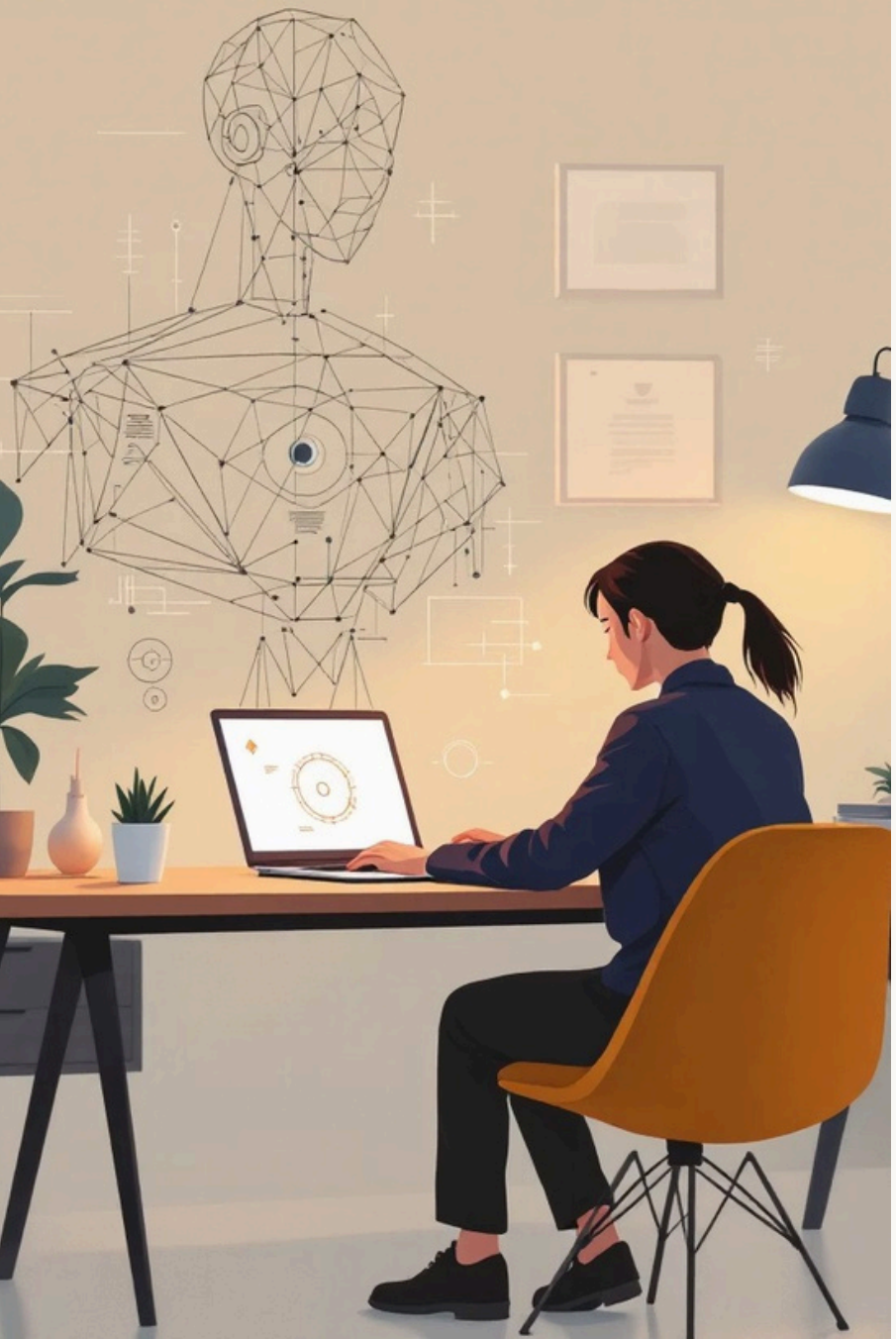

Agentic AI-Driven Online Proctoring System

Anautonomous, real-time proctoringsolutionthat ensuresacademic integritythrough intelligent perception,reasoning, andautonomous action

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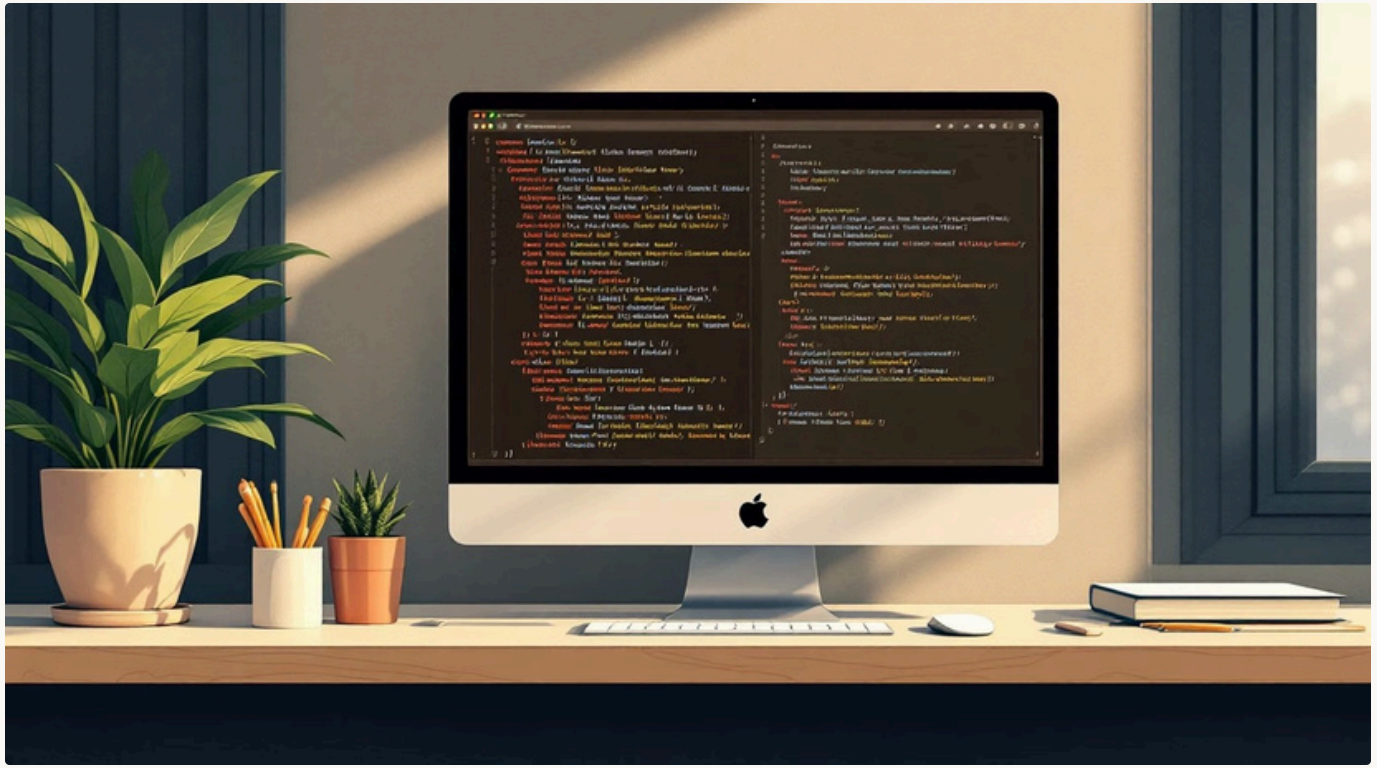
Project Overview



This project represents a fundamental shift from traditional proctoring software that merely records footage to an **autonomous intelligent system** capable of real-time decision-making.

Built using Python with **OpenCV** and **MediaPipe** for high-performance facial landmark tracking, the system processes 468 3D facial landmarks at 30 FPS on standard consumer hardware.

The architecture combines computer vision with temporal reasoning to create a lightweight, scalable solution suitable for mass deployment across educational institutions.

Four-Agent Architecture

The system operates through a continuous loop of four specialized AI modules working in concert

Vision Agent

Perception and facial landmark tracking

Memory Agent

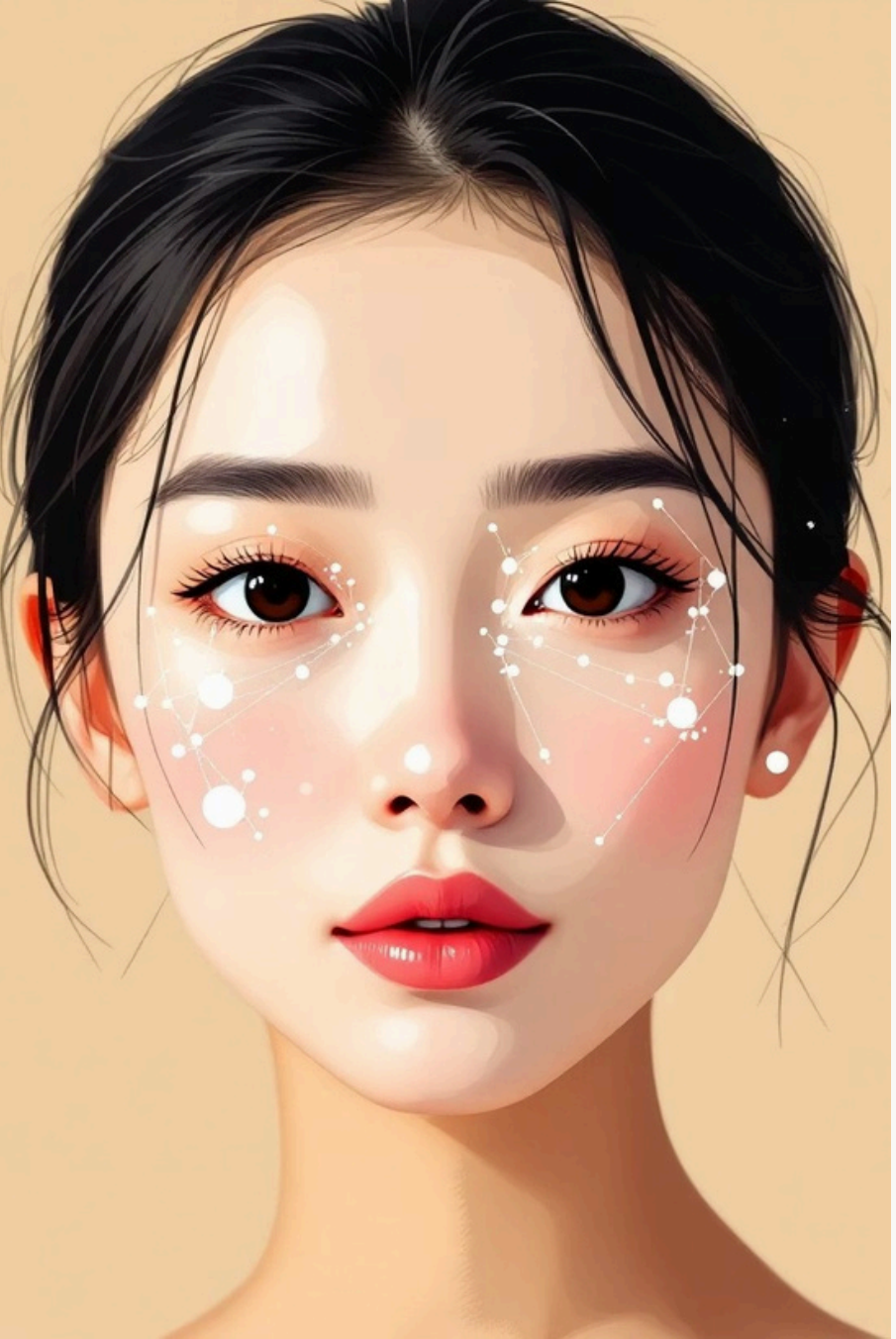
Temporal state tracking

Reasoning Agent

Risk assessment and explanation

Action Agent

Enforcement protocol



Vision Agent: Perception Module

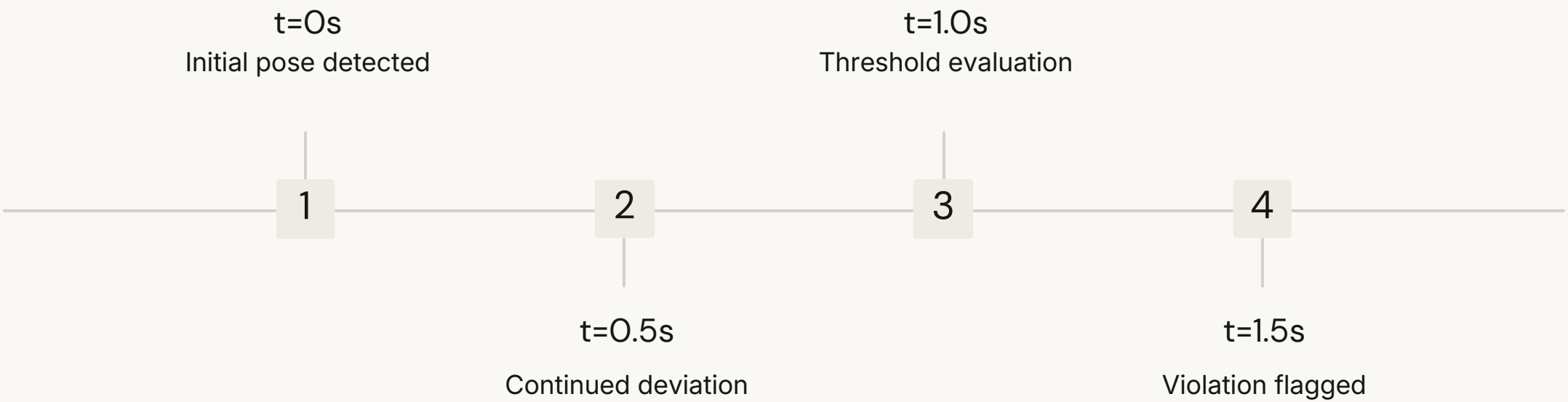
Core Capabilities

- 4683D facial landmarks tracked at 30 FPS
- Multi-person detection in frame
- Head pose estimation with 6 degrees of freedom
- Real-time gaze direction analysis

TechnicalImplementation

MediaPipeFace Meshprovides sub-millimeter accuracy. Head-turn thresholds calculated as percentage of face width for scale invariance.

Memory Agent: Temporal State Tracking



The Memory Agent maintains a rolling buffer of the past 2 seconds of candidate states. This temporal reasoning ensures the system distinguishes between accidental movements (sneezing, involuntary glances) and sustained suspicious behavior. Only deviations persisting beyond 1.5 seconds trigger warnings.

Reasoning & Generative Agent

Human-Readable Explanations

Generates detailed risk assessments for every flagged behavior, explaining *why* a movement triggered suspicion.

Context-Aware Evaluation

- Analyzes behavioral patterns
- Weights risk factors
- Provides audit-ready documentation

Action Agent: Enforcement Protocol

01

Level 1: Auditory Warning

Single beep alerts candidate of suspicious behavior

02

Level 2: Integrity Breach

Second violation triggers automatic exam termination

03

Level 3: Evidence Generation

CSV log created with timestamps and violation details

Key Technical Innovations

Temporal Reasoning Buffer

1.5-second persistence threshold filters noise while catching sustained violations

Scale-Invariant Logic

Head-turn thresholds calculated as face-width percentage, ensuring accuracy regardless of webcam distance

Evidence Log Generation

Automated CSV audit trail with timestamps, violation types, and risk levels

Evidence Log & Data Integrity

Transparent Audit Trail

Everysession generates a comprehensive CSV log containing:

- Precise timestamps for all events
- Violation type classification
- Risk level scoring (1-5 scale)
- Generative AI explanations
- Raw facial landmark coordinates

Provides educational institutions with complete transparency and defensible evidence for academic integrity decisions.

Transforming Academic Proctoring

From Passive to Active

Replaces recording-only systems with intelligent, autonomous enforcement

Fair & Transparent

Scale-invariantlogicand temporal reasoning reduce false positives while maintaining rigor

Accessible & Scalable

Runson standardconsumerhardware without cloud dependency

By combining Generative AI explanations with autonomous decision-making, this project bridges the gap between AI detection and administrative needs—creating an intelligent, fair, and auditable enforcement system for modern education.

Thank You

For yourtime andconsideration. I believe this system offers a robust, fair, and scalable solutionformaintaining academic integrity in online environments.